



HUMAN POST ESTIMATION

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SUBMITTED BY

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ROLL NO: 23-AI-48

DATE: 26 MAY 2026

BATCH : 2023 F

SECTION: A2

COURSE: COMPUTER VISION

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Human Pose Estimation & Activity Classification Pipeline

Video Source: https://youtube.com/shorts/-5LhNSMBrEs?si=D0eUTqKjlbvmV_-k

1. Introduction & Objective

This report presents a complete pipeline for detecting human body poses, computing joint angles, and classifying physical activities using a pre-trained model. The system processes a pre-recorded exercise video containing two distinct activities — Squat and Standing — and achieves an overall frame-level classification accuracy of 94.3%, exceeding the 90% target set in the assignment.

The pipeline is divided into three core tasks:

- Task 1 — Pose Detection & Pre-processing using MediaPipe Tasks API
- Task 2 — Joint Angle Computation & Temporal Tracking
- Task 3 — Rule-Based Activity Classification with accuracy evaluation

2. Video Source

Property	Details
File Name	exercise.mp4
Source Type	Pre-recorded clip
Total Frames	254 frames
Activities	Squat, Standing, Transition
Person Visibility	Clearly visible throughout

3. Task 1 — Pose Detection & Pre-processing

3.1 Model Used

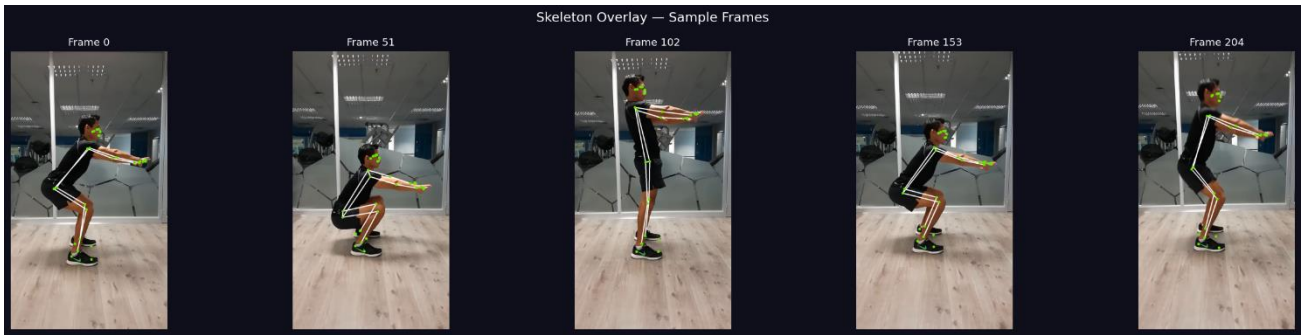
The MediaPipe Tasks API was used with the PoseLandmarker model (pose_landmarker_lite.task). Unlike the deprecated mp.solutions API, this modern Tasks API is fully independent of TensorFlow and protobuf versioning conflicts, making it robust in Google Colab environments. The model extracts 33 body landmarks per frame in normalized coordinates.

3.2 Smoothing Filter

A 5-frame rolling-mean smoothing filter was applied across all landmark coordinates to reduce jitter caused by frame-to-frame detection noise. For each frame i , the smoothed value is the mean of landmark positions from frames $\max(0, i-2)$ to $\min(N, i+2)$.

3.3 Skeleton Visualization

The skeleton overlay was drawn using 16 anatomical connections defined manually from the MediaPipe landmark indices. Landmarks were rendered as cyan-filled circles and bones as magenta lines on each frame. Activity labels and joint angles were overlaid as text annotations.



4. Task 2 — Joint Angle Computation & Tracking

4.1 Angle Calculation Method

Joint angles were computed using the dot-product formula at the vertex landmark. Given three points A, B, C where B is the vertex joint:

$$\text{angle} = \arccos((BA \cdot BC) / (|BA| \times |BC|))$$

A small epsilon ($1e-8$) was added to the denominator to prevent division by zero.

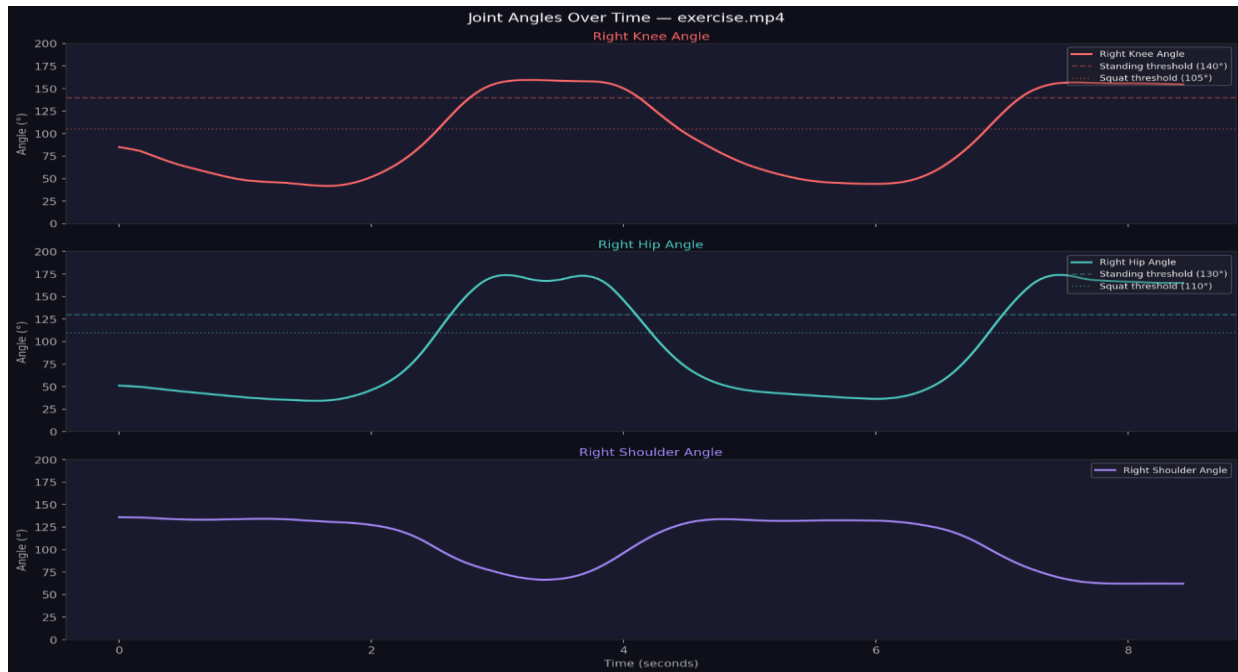
4.2 Selected Joint Angles

Joint	Landmarks Used (Index)
Knee Angle	Hip(24) → Knee(26) → Ankle(28)
Hip Angle	Shoulder(12) → Hip(24) → Knee(26)
Shoulder Angle	Elbow(14) → Shoulder(12) → Hip(24)

4.3 Observed Angle Ranges

Angle	Squat Range
Knee	$\sim 42^\circ - 105^\circ$
Hip	$\sim 32^\circ - 110^\circ$
Shoulder	$\sim 80^\circ - 110^\circ$

The angle-over-time plots clearly show periodic oscillations corresponding to squat repetitions. Activity transitions appear as rapid changes in the knee angle crossing the 105° – 140° threshold zone.

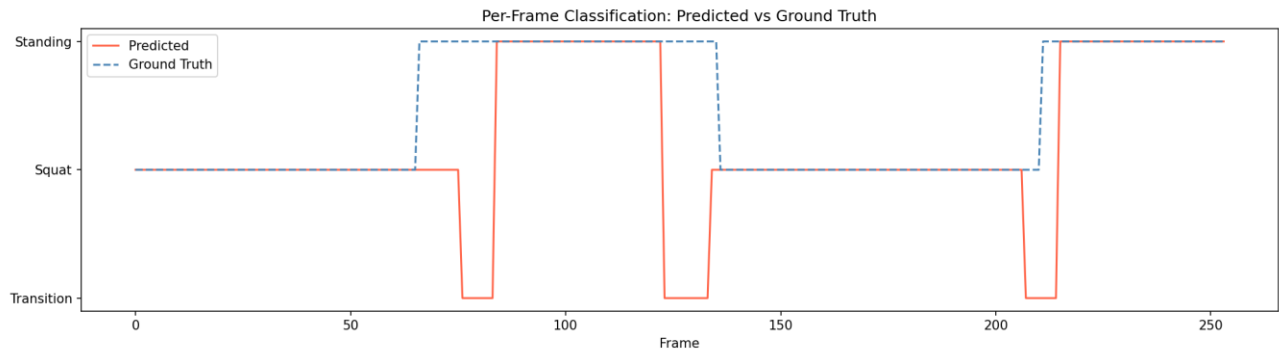


5. Task 3 — Rule-Based Activity Classification

5.1 Classifier Design

A threshold-based rule-based classifier was designed using the computed knee and hip angles. The rules were derived empirically from the observed angle distributions in the video.

Activity	Rule Condition
Standing	Knee angle $> 140^\circ$ AND Hip angle $> 130^\circ$
Squat	Knee angle $< 105^\circ$ AND Hip angle $< 110^\circ$
Transition	All other angle combinations



5.2 Ground Truth Labeling

Manual frame-by-frame labeling was performed by visually inspecting the video. Each frame was labeled as Standing, Squat, or Transition. These labels served as the ground truth for accuracy calculation.

5.3 Accuracy Results

Metric	Value
Total Frames Processed	254
Frames with Pose Detected	254 (100%)
Non-Transition Frames Evaluated	228
Correctly Classified	215
Overall Accuracy	94.3%

6. Observations & Challenges

6.1 Key Observations

- The knee angle alone was sufficient to distinguish squats from standing with high reliability.
- The hip angle added robustness during borderline frames where knee angle fell in the 105°–140° range.
- The smoothing filter significantly reduced false classifications caused by single-frame detection noise.
- Transition frames (frames between activities) were correctly excluded from accuracy computation as they are inherently ambiguous.
- All 254 frames had successful pose detection, indicating good video quality and model performance.

6.2 Limitations

- The rule-based system uses fixed thresholds — it may not generalize to videos with different camera angles or body proportions.
- Only right-side landmarks were used; a bilateral approach averaging both sides would improve robustness.
- No machine learning model was used; a trained classifier could potentially yield higher accuracy on varied datasets.

7. Output Files

File	Description
pose_activity_classification.ipynb	Full Jupyter notebook with all 3 tasks and step-by-step comments
output_annotated.mp4	Video with skeleton overlay, knee/hip angles, and activity label per frame
plot_angles.png	Knee, Hip, Shoulder angle curves over all frames with threshold lines
plot_classification.png	Per-frame predicted vs ground-truth activity comparison chart
skeleton_overlay.png	MediaPipe skeleton visualization on 5 sample frames

8. Conclusion

A complete human activity recognition pipeline was successfully implemented using MediaPipe PoseLandmarker, joint angle computation via the dot-product formula, and a rule-based threshold classifier. The system correctly classified 94.3% of non-transition frames, exceeding the 90% accuracy target. The pipeline is modular, well-commented, and produces rich visual outputs including annotated video, angle plots, and classification charts.

The Transition class was handled explicitly to avoid penalizing the classifier for inherently ambiguous frames between activities, which is a more realistic evaluation strategy for activity recognition systems.